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Formal Models in Comparative Politics & International Relations

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Abstract

Standard game-theoretic accounts often rationalize departures from fully rational play by appealing to exogenous "shocks" or moments of surprise. Yet many strategic encounters suggest that what matters is not merely an external disturbance but an endogenous shift that reshapes perception, signalling, and restraint. This paper develops an illustrative framework using two elephant-human encounters reported by Safina (2015) as motivating cases. In one episode, an elephant charges a tourist yet brakes before contact; in another an elephant injures a shepherd and subsequently guards him until help arrives. Canonical conflict models can capture incentives and credible threats in both interactions, but they struggle to accommodate the joint pattern "attack, then care." I treat the two episodes as potential outcomes of a common underlying interaction, and use a causal directed acyclic graph to make explicit the missing variables that standard models implicitly omit. I argue that surprise operates as an informational shock that can trigger aggression, while empathy functions as an affective regulator that modulates signal intensity and can generate post-conflict assistance. To formalise this intuition, I sketch an adaptation of Groseclose and Snyder's (1996) vote-buying game, in which competing emotions allocate limited intensity across competing "impulses" to determine behaviour. The resulting perspective offers a tractable route to richer models of bounded rationality and yields testable implications about when shocks produce escalation versus restraint.

1 Introduction

In this paper I wish to describe an intuition regarding the way game thoery has yet to formalise certain interactions between agents. Game thoerists consider surprises and shocks as the rationale for apparently irrational behaviour. I believe that the following paragraphs will convice the audience of the value of this contribution for two reasons; first, I hope the description provided will be used and applied further. Second, the conjecture about the role of emotions adds a new perspective to the literature about bounded rationality.

The intuition concerns animal behaivour that has drawn heavily on game thoery. Although it may seem distant from standard economic applications, still, it deals with the problems that game thoerists often face: the first is to represent unobserved mental processes, this means taking into consideration and inferring unobserved variables. The second problem deals with the creation of a formal idea about these processes. The last problem connects the formal idea to observed behaviour and derives an equilibrium, thus a state of the world with particular properties. I aim to show that seeking an explanation outside the mind is unnecessary, since it may well lie within it.

To streamline the exposition and to make the model more directly applicable, I exploit a singular situation described in Safina (2015). This book is an amazing challenge for every person who has ever had to face a game-theoretic exercise. To support his thesis, the author provides the reader with detailed descriptions of animal interactions. All species selected belong to the category of "cognitively sophisticated" animals, animals that are capable of emotions and complex cognitive processes. I believe that each of these interactions is a useful game-theoretic exercise, since they force game thoerists to apply its knowledge out of his area of expertise.

This book also delivers a straightforward setting for these exercises, since animals offer a simplified set of agents for game theory. They are complex enough to beign represented as rational players, but they are not as complex as humans, so that most of the assumptions and simplifications have more grasp into reality. In this respect, elephants appear to be a particularly apt exercise. There is plenty of documentation about their behaviours, and there is plenty of evidence that they act in accordance with a *mind*. According to Safina (2015), they are almost humans, and they often interact with humans.

I will formalise my conjecture through a Causal Directed Acyclic Graph (DAG). Since I wish to norrow down the gap between thoery and practice, I believe this tool is well suited to the research design. Additionally, it provides a more transparent version of the analysis, clarifiyng the variables of interest and the limits of the assumptions.

The final aim of this article is to clarify a contradiction between two similar situations that game theory formalises in different ways.

1.1 Beyond Words

Carl Safina is a biologist and has written a widely read essay (Safina, 2015) about animals' minds. He argues that until now, biologists have been reluctant to deepen the study of animals' minds and "souls" and that is high time they did so. The curious thing about this book is that, in doing so, he draws readers' attention to a problem closely related to game theory. He tries to rationalise the conduct of some animals, arguing that their behaviour may be explained if we accept they are "intelligent", meaning that a mind guides their actions and that they are sensitive to emotions. This book is an attempt to place animals' minds at the same level of human minds. I hope the animal-world examples will:

- change the way we theorise a broader set of agents and interactions.
- be used to test the claims and strengthen the empirical support.
- motivate a refined equilibrium concept.

1.2 The Problem

This section aims to clarify the setting captured by the game-theoretic representations, and to make the reader aware of the basic intuition. In the examples Safina describes in his book (Safina, 2015) animals seem to be driven by rationality and emotions, much like humans. *These are brief summaries of the two episodes.* Both interactions between humans and elephants happened in a wildlife park in Africa. In both, the surrounding environment appears similar.

Case $\mathbf{n}^0 1$

A staff member of a wildlife park in Africa reports an "attack" by an elephant on a tourist. The scene clearly highlights the signals that the elephant sends to the tourist. It also clearly identifies the reasons why it is logical to believe that the elephant intentionally avoided hurting the tourist. In particular, the staff member reports signs on the ground that strongly suggest the intention to halt the charge against the tourist moments before impact. We can formalise this game through a deterrence in which the elephant sends a threatening signal without intending to harm him.

Another situation, recorded in the same environment, shows how similar conditions can produce an opposite pattern of actions.

Case $\mathbf{n}^0 2$

In the same wildlife park in Africa, a rescue group was dispatched to reach a shepherd who was badly hurt by an elephant. When the staff approached him, they reported that the shepherd stopped them from harming the animal. He argued the elephant had kept him safe during the night. From the conduct of the animal it seemed clear what the elephant intended. After the attack, it soon realised it had misinterpreted the signals from the shepherd, and after hurting

him, the animal took care of him until the rescue group reached the place of the accident. Here, the same signalling logic as in the first example breaks down: the equilibrium is reached, yet the elephant attacks. However, the event suggests a richer model of rationality, where the animal is able to recognise an "error".

Both scenarios can be framed within game theory, yet each reveals elements of cognition and emotion that standard frameworks overlook.

In particlar, we consider three puzzles that emerge from these examples:

1. Why does the animal attacks and then provides assistance? Isn't it contradictory?
2. Why does the animal initiates an attack only in one of the two cases?
3. What is rational? To attack and regret the decision, or not attack at all? With respect to what criteria do we define which is the best strategy?
4. Which is the best strategy in evolutionary terms?

1.3 Parallel with Game Theory

I believe game-theoretic analysis accurately captures the key strategic features of both cases. For example, the fact that the animal initiates a conflict with humans and tends to be pacific with most of other creatures is due to the fact that humans are one of the few threats they face in nature. Game theory correctly models agents as driven by incentives, yet it does not account for interactions in which the agent acts violently because of *its malicious nature*. Similarly, game theory formalises credible threats as equilibria depending on the information available to players: not the turist nor the sheperd intended to intentionally harm an animal that could plausibly kill them, for example.

Additionally, there are plenty of models that suit each of the two situations. For instance, as shown below, some formalisations explain the behaviuor of the first case thorugh the concept of mimicry: the elphant knows it will not cause harm to the tourist. However, the turist cannot distinguish between an animal with good or bad intentions, and the elephant exploits this weakeness for defensive porposes.

Thus, both animals and humans may be depicted as rational agents, since their behaviour appears to be consistent with consistent with the incentives they face and with a broader objective, if not with a specific aim in mind. Both appear to be able to communicate with each other, indicating that both species are adapting to signals from other agents and that they can act as informed agents. And both seem to be aware of each other's strengths, at least partially (Schelling, 1960; J. M. Smith, 1973). Again, in both encounters, the animal appears peaceful and willing to avoid or prevent conflict, at least up to a point.

1.4 Representative Agents

In both cases analogies between humans' and animals' minds do not appear to be straightforward. However, most of their differences need not be represented in game theory formalisations. Additionally, game theory is often applied at different levels of analysis (e.g. nations' behaviour), which means that only the crucial features of agents need to be represented.

Thus, under the descriptive limits of these games and under their assumptions, I believe no confusion arises when I use "human", "elephant", and "agent" interchangeably. *I will differentiate between these agents only in those cases in which it is necessary, and with due warning to the reader.*

2 Causal Inference

The discussion so far motivates a comparison to frameworks that model how unobserved factors influence observed outcomes. Causal inference (Pearl, 2009) suggests that in order to answer the question of interest, we may also think of an unobserved confounder that that is mistakenly excluded from the framework. I argue that an omitted variable is present in this setting.

2.1 Potential Outcomes Framework

This section formalises the intuition using a DAG. To make the subsequent discussion accessible, it first outlines how the potential-outcomes framework applies to this set-up.

A potential outcome is a counterfactual. Thus, it is a scenario that we do not observe; we think of it as the closest plausible scenario that might have happened had the actual one not occurred.

In the case of the two interactions presented here, each of them may be considered the counterfactual scenario of the other. Since counterfactual outcomes do not exist, we use real and observed cases to mimic them. The underlieng belief is that no researcher is as informative as reality itself in producing examples of what might have happened. If we rely on observation, the role of the researcher is limited to selecting the most appropriate counterfactual from those offered by the natural world.

Hypothesis: absent any causal effect of emotions, the instance involving the elephant and the sheperd would have been identical to the situation offered by the other elephant and the tourist.

2.2 Application of the Potential Outcomes Framework

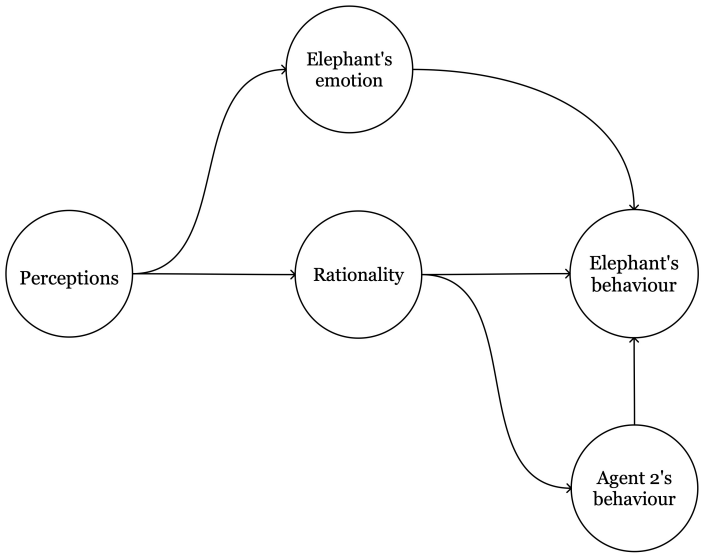

Figura 1: *Causal DAG.*

The graph shows five circles that represent our variables. We temporarily set aside the emotion node to describe the baseline case.

In the baseline set-up agents are represented by perceptions, rationality and behaviour. The former two shape the latter, which is the outcome variable of interest. A change in perceptions corresponds to a change in actions. Rationality is modelled as the decision rule or the algorithm agents have in their minds. It is the function that represents utility. Two players with the same utility function act in the same way when facing the same game. This means that a change in rationality corresponds to a chenge in behaviour.

Our outcome variable is the only one we can observe and measure. In the two occurrences described by Safina (2015), we *observe* the conduct of elephants and humans. However, we can only *infer* rationality and perceptions.

When we represent two agents using three variables, we make use of three circles for each player. Nonetheless, the figure presents only one circle for perceptions and only one circle for rationality.

This choice reflects a simplification. In order to ease the analysis, we assume that both agents perceive reality in the same way. Thus, both receive the same stimuli from the environment. We also assume that they process these stimuli through the same algorithm. They weigh the same benefits and costs in the same way. These assumptions symplify the baseline setting because restrict the set of potential outcomes: we consider only two possible outcomes: rationality in state

A or B and perceptions in state A or B.

How do we distinguish between the two agents in the Causal DAG? They differ in the realised behavioural outcome. While human conduct is constant, elephant is different in the two instances (The Problem), this is why we have one circle for each player.

We have 4 circles, three of which are constant in the two occurrences. The fifth one represents elephant's emotions. This variable provides the key source of variation across cases. Thus, it can explain the change in elephant behaviour.

2.3 The Rationale for using Causal Inference
From the General Case to a Single Case–and Back Again

von Neumann and Morgenstern (1990, chap. 2, p. 14) write: "An almost exact theory of a gas, containing about 10^{25} freely moving particles, is incomparably easier than that of the solar system, made up of 9 major bodies; and still more than that of a multiple star of three or four objects of about the same size. This is, of course, due to the excellent possibility of applying the laws of statistics and probabilities in the first case. [...] The problem must be formulated, solved and understood for small numbers of participants before anything can be proved about the changes of its character in any limiting case of large numbers".

To pursue this suggestion, we move the abstract general case of rational-agent model to analyse specific interactions in which the representation fails to adequately map reality.

We then turn to causal inference in order to reconstruct a more general account: namely, to move from the idiosyncrasies of a single interaction to a *new* general framework. This requires abstracting away from irrelevant features and identifying a shared causal mechanism. The result is a revised pattern: a new map that can guide inference, organise the evidence, capture the data-generating process.

3 Models

We revisit the four puzzels raised by the animal examples in subsection 1.2 (The Problem):

1. Why does the animal initiates attack and then provides assistance? Isn't it contradictory?

2. Why does the animal initiates attack only in one of the two instances?

3. What is rational? To attack and regret that decision, or not attack at all? With respect to what criteria do we define which is the best strategy?

4. Which is the best strategy in evolutionary terms?

The following section draws on two classical thoeries of games with the aim of answering the four puzzels. The explanations improve upon these theories and support new idea.

3.1 Model n^{0}1: Surprise as Shock

The first two classical theories we can appeal to are *deterrence* and *bargaining*. Reiter (2003) summarises these positions arguing that: "Classic deterrence theory predicts that as an imbalance of power between two states grows, war becomes more probable. The bargaining model, *on the other hand*, argues that it is not an imbalance of power that causes war, but rather disagreement over the balance of power (and over the likely outcome of war)"*(Emphasis added.)*

In the light of formal representations of conflic, the first question can be rephrased as follows: can we depict Case n^{0}1 as a deterrence game and Case n^{0}2 as a bargaining game?

Three doubts naturally arise as obstacles to the adoption of this view. First, there is no intelligible reason why the animal acts through deterrence in one occasion and thorugh bargaining in the other. Secondly, we desperately need a justification for our distiction. Otherwise, We would incur in the unplesaint situiation of modelling two identical interactions through different formalisation without any specific reason. Supposing, we are able to find thoeritic support for these puzzles, we would still need to explain why the animal regrets the attack. Therefore, the third doubt arises because it appears unlikely to explain why the animal regrets the attack by making a distinction between deterrence and bargaining.

The decisive, and *omitted*, variable could be the element of surprise. Fearon (1995) argues that, *if we assume that conflicts are costly*, no war is possible when the outcome is clear to both sides. However, if we add surprise to Fearon (1995)'s framework, we can derive an equilibrium. The two versions of the elephant-human encounter are similar in strategic structure. In the tourist scenario, there is no surprise: the elephant is aware that humans might approach and disturb it, and it knows where to find them.

The situation involving the shepherd, however, contains the element of surprise, which may justify a different behavioural response.

In order to respect time and pages constraint, the main idea presented here cannot be further developed. The reason behind this choice is to prioritise the model that is introduced in the next section.

3.2 Model n^{0}2: Vote Buying

3.2.1 Back to Groseclose and Snyder (1996)

This section develops I propose a reformulation of a well-known formalisation by Groseclose and Snyder (1996). I intend to draw on the equilibrium the authors

derive, to support the argument developed in the DAG by providing an additional illustration. In doing so, I aim to improve upon the previous model (See subsection 3.1: Model n^{0}1: Surprise as Shock).

In their article, Groseclose and Snyder (1996) explain why, in the US Congress, the number of votes often exceedes the minimum required to pass a bill. They argued that this is due to the strategic interaction between lobbying groups competing for support on different bills. These groups *bribe* members of Congress to convince them.

The central idea in the paper is illustrated by:

"The legislature contains seven members, all indifferent between the two proposals, that is, $v(i) = 0$ for all i. Since B will move second and attack the weakest part of A's coalition, A will offer the same bribe, a, to all legislators he bribes. Let $m + 4 \geq 0$ be the size of the coalition A bribes. If $m = 0$, then B can defeat x by paying $a + \varepsilon$ to one member of A's coalition. Thus, A must set $a \geq W_B$ to win, and A's total bribes are then $4W_B$. If $m = 1$, then B must pay $a + \varepsilon$ to two members of A's coalition to defeat x. Thus, A must set $a \geq W_B/2$ to win, and his total bribes are then $5W_B/2$. This is less than $4W_B$, so A is better off buying a supermajority of size 5 than a minimal winning coalition. If $m = 2$, then B must pay $a + \varepsilon$ to three members of A's coalition, so A must set $a \geq W_B/3$ to win, and his total bribes are then $2W_B$. In fact, A minimizes his total payments by bribing all seven legislators. Here, $m = 3$, and A's total bribes are $7W_B/4$. Clearly, there is an equilibrium where x wins if and only if $W_A \geq 7W_B/4$. If $W_A < 7W_B/4$, then there are only equilibria in which no bribes are paid and s wins." (Groseclose & Snyder, 1996)

3.2.2 Representation through Vote Buying

Consider a council composed of several members. These members are non-strategic players whose votes can be influenced. There are two active players, conceptualised as two groups, each seeking to "buy" the votes of the council members in order to determine the policy agenda. Each group possesses a budget, and bills are voted on by the council according to a pre-defined rule, assumed here to be the simple majority rule. This setup corresponds to the classical *vote-buying game* as formalised by (Groseclose & Snyder, 1996).

The objective of this analysis is to adapt this framework to the so-called *elephant-shaped* case. This adaptation allows the original model's equilibrium structure to be embedded in a cognitive-emotional set-up.

- Let K denote the total number of *thoughts*.
- The two players represent *opposing emotions* that govern the internal dynamics of the elephant's cognition. "Setting the agenda" refers to determining the behaviour.

- Each player's *budget* corresponds to the *intensity* of its emotion. A higher emotional intensity enables the emotion to "purchase" (secure) a larger number of thoughts.

Notation:

Players: $N = \{X, Y\}$ or $N = \{Emotion1, Emotion2\}$.

Histories: $H = \{\varnothing, \{x\}, \{x, y\}\}$. Wwere x and y indicate payment vectors in the baseline setting. Emotions use units of payment (emotional intensity) to corrupt a thought (or a judgement).

Player function: $p(\varnothing) = X; \quad p(x) = Y$

Actions: $A(\varnothing) = \{x\}; \quad A(x) = \{y\} \; \forall \, x$.

Each player *receives* V_i if the elephant acts accordingly.

Each player loses $\sum_{j=1}^{k} x_j$ for each unit of intensity spent.

This adapted framework enables a formal representation of conflict as a strategic interaction between emotions competing for cognitive control. Each emotion allocates its limited intensity across thoughts, attempting to secure a majority sufficient to determine the elephant's conduct choice. The resulting equilibrium can be interpreted as a stable emotional configuration in which neither emotion can improve its outcome given the other's allocation strategy. Let μ denote the minimal number of thoughts required to *set the agenda*, that is, to achieve behavioural dominance of one emotion over the other. Then:

$$\mu = \frac{k+1}{2}$$

4 Literature Review and Limitations

4.1 A Review of Models of Conflicts

Further exploration of the conflict-games literature may either support or qualify the analysis presented so far. In any case, already-established models provide a useful point of comparison. Some of the papers that stand out include Schelling (1960, 1966). On the other hand, it may be of interest to focus on McKelvey and Palfrey (1995) to further develop the design.

4.1.1 A Review of Models of Conflicts with a Focus on Bargaining

One prominent strand of the literature examines on bargaining models of conflict. Key contributions to this field number Fearon (1995, 1996, 1998) and Schelling (1960, 1966). Reiter (2003) offers an in-depth review of the field. He provides an historical perspective and discusses more recent works. Additionally, he summarises the most important ideas.

Reiter (2003) says: "James Fearon prominently contributed to this scholarship with his 1995 paper "Rationalist Explanations for War." He usefully highlighted

the point that if states agreed on the outcome of a possible war, they could probably avoid war. Central to Fearon's paper is the importance of focusing on states' disagreement over their capabilities and/or resolve. [...] Fearon uses a bargaining model to argue that if two states in dispute know the outcome of a possible war, they should in general prefer to reach a deal that would reflect the hypothetical postwar political settlement, rather than fight, reach that same settlement, and also suffer the costs of war. This view assumes that fighting itself is costly–that the belligerent always suffers some negative utility, no matter how the issue at stake is settled. In fact, this assumption becomes necessary when one asserts that two states would always prefer to reach a bargain without fighting rather than fight and then reach the same bargain. Within his bargaining model, Fearon develops three conditions under which war is possible."

To broaden the overview, some other salient articles comprise Filson and Werner (2004), Kirshner (2000), A. Smith and Stam (2004), Wagner (2000), and Wittman (1979).

4.2 Limits of Theoretical Models

This discussion is necessarily selective and cannot provide a comprehensive assessment of the formal-modelling literature. In the previous pages I have presented a personal re-elaboration along with a brief summary of the relevant literature in this area. This is certainly not enough to assess the state of the models of this essay. However, the exposition I provide should not be taken as comprehensive, since there is no such thing as *a single correct or definitive modelling framework.*

In this respect, it is possible to further expand the discussion on limitations. The literature offers some very valuable contributions, among others, I discuss: Ashworth et al. (2021), Bräuninger and Swalve (2020), and Pareto (1916). These essays presents different ways to combine empirical and theoretical researc together with some insightful interpretations of the meaning and scope of models.

4.3 Connection between Game Theory and Biology

There are many important applications of game theory in biology. Particularly influential examples include: Axelrod and Hamilton (1981) and J. M. Smith (1973). Future work may shed light on more recent and equally notable applications. A vast literature covers the topic of evolutionary games as well: Hofbauer and Sigmund (2003), Kandori et al. (1993), Nowak (2006), and Taylor and Jonker (1978). In particular, Hofbauer and Sigmund (2003, chap 12) offers a clear treatise of powerful analytical tools together with applications of these game-theoretic tools in biology.

Unfortunately, a more detailed discussion of these works must be postponed.

Conclusion

The puzzle motivating this paper was deliberately simple: two elephant-human encounters occur in broadly similar environments, yet they generate sharply dif-

ferent behavioural trajectories. In one occasion, the elephant's threat is forceful but calibrated; in the other, violence is followed by a striking form of care.

Standard signalling logic is largely successful in explaining why an elephant might issue a credible threat and why a human might respond cautiously. What it does not readily explain is the combination of escalation and subsequent protection–an outcome that looks puzzling from the standpoint of canonical rational-agent models. The central claim advanced here is that the apparent inconsistency largely reflects omitted elements of the model: the relevant "shock" can be internal, and the relevant constraint can be affective.

The causal-inference lens is useful. Treating the two episodes as potential outcomes of a shared underlying interaction makes the research design explicit: the cases serve as mutual counterfactuals, and the goal is to isolate which latent factor, omitted in standard game-theoretic frameworks, generates the divergence. Within this framing, surprise functions as an informational discontinuity that plausibly shifts the elephant's perception of threat and thus the equilibrium-relevant mapping from signals to actions. Crucially, however, surprise alone cannot account for the post-attack behaviour. That second step motivates the introduction of empathy as a regulator: a force that can attenuate aggression, reshape the "intensity" of signals, and generate assistance once the situation is reinterpreted.

This conceptual move has a representational payoff. Instead of treating emotions as noise that perturbs otherwise coherent strategic choice, the paper sketches a way to represent emotions as structured, competing influences over cognition. The adaptation of the vote-buying framework provides a particularly transparent metaphor: thoughts play the role of passive "voters", while emotions act as active players allocating scarce intensity to secure agenda-setting power–here, the determination of behaviour.

In equilibrium, the elephant's observable trajectory corresponds to a stable emotional configuration: neither emotion can improve its outcome given the other's allocation. Read in this way, "attack, then care" is not a contradiction so much as the observable consequence of a reallocation of cognitive control after a misinterpretation is recognised.

The broader implication is that many phenomena labeled as "shocks" in political science and international relations may be mischaracterised when treated as purely exogenous disturbances. A framework that makes these mechanisms explicit can therefore complement familiar strategic models, not by abandoning rationality, but by widening the set of state variables that rational choice is allowed to condition on.

At the same time, the limits of the present contribution should be stated plainly. A fuller treatment would require specifying payoffs, information structures, and equilibrium selection more rigorously. Future work could therefore proceed by tightening the formal model, including the relationship between sur-

prise and empathy as distinct but interacting state variables.

The main message is modest: when game theory explains puzzling conduct by invoking "surprise", some times it may be pointing towards a missing internal mechanism. Making that mechanism explicit–rather than treating it as residual noise–can turn an ad hoc rationalisation into a tractable extension of formal modelling, one that is better aligned with the texture of real strategic behaviour.

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